TEJAS L

Aerospace Engineering Student, RV College of Engineering

Bengaluru, India · +91 7892467274 · 050tejasl@gmail.com · linkedin.com/in/tejas-l · github.com/tejasl-portfolio

SUMMARY

Aerospace Engineering student (CGPA 8.80) with hands-on experience across propulsion design, CFD/FEA, orbital mechanics, embedded systems, and controls — carrying projects from analytical design through simulation to manufactured, physically tested hardware. Comfortable presenting technical work to review panels and coordinating across cross-functional sub-teams.

EDUCATION

RV College of Engineering — B.E., Aerospace Engineering *Bengaluru, India*

CGPA: 8.80 / 10 (through 6th semester) · Expected Graduation: 2027

EXPERIENCE

Aero-Propulsion Engineering Intern — Prime Tooling *Jul 2025 – Sep 2025*

– Designed a supersonic nozzle profile using the Method of Characteristics (MOC) to achieve shock-free, isentropic flow expansion

– Reverse-engineered the mechanical design and actuation logic of a missile fin control system, applying DFM principles across the CAD/CFD/FEA workflow

Engine Sub-system Member — Ashwa Mobility Foundation (RVCE Formula Student Team)

– Designed and manufactured the endurance-run fuel tank and dry-sump oil tank for the team's combustion FSAE car, cross-validating fuel consumption sizing across two independent methods to 1.55% error

– Represented the Powertrain sub-team at the Engineering Design Presentation (EDP) event, Formula Bharat 2026, presenting the tank redesign to a judging panel — placed 5th nationally

PROJECTS

Energy-Aware Digital Twin for LEO Satellite Constellations *RVCE Interdisciplinary Project, AY 2025–26*

– Designed a 5-satellite STK scenario and built an SGP4-to-HPOP propagation pipeline (EGM2008 gravity, Jacchia-Roberts drag, solar/lunar third-body effects); ran AdvCAT3 conjunction screening over a 24-hour window, flagging 50+ close-approach events (min. separation 6.1 km)

– Fed physics-validated orbital data into a teammate-built GNN/RL collision-avoidance system that achieved a 92% collision-risk reduction and 100% satellite survival rate

Fuel & Dry-Sump Oil Tank Design — Formula Student Powertrain *Individual Project, Ashwa Mobility Foundation*

– Cross-validated endurance-run fuel consumption via two independent methods (MoTec log-based duty-cycle mapping and OptimumLap simulation) to 1.55% error; sized a new 7.15 L tank with an asymmetric baffle and ~450 ml anti-starvation reservoir, validated via multiphase VOF CFD slosh simulation at 1.358g cornering load

– Redesigned the dry-sump oil tank (3.32 L usable, 0.82 L above requirement) and carried both parts to manufacture in welded Aluminium 6061, including a machined sealing flange sized to Parker O-Ring Handbook tolerances

Camshaft Meshing Automation in ANSA *AS244AI Course Project, Even Sem 2024–25*

– Built the SolidWorks CAD model and wrote the Phase 1 validation script for an object-oriented Python pipeline automating camshaft meshing in BETA CAE ANSA; tuned NASTRAN mesh-quality thresholds for a rotating-machinery component

- Replaced a manual, 10+ click ANSA workflow with a batchable, fault-tolerant pipeline across 3 element sizes, cutting a full convergence study to ~11 seconds and producing a 98.9% hexahedral mesh with zero inverted elements

CFD Study: Effect of Surface Roughness on Oblique Shock Behaviour *Gas Dynamics AS362IA, AY 2025–26*

- Validated a RANS/SST k-ω solver (ANSYS Fluent) against a published rough-wall supersonic benchmark (0.57% error) before studying a Mach 3 double-wedge model
- Swept equivalent sand-grain roughness height (0–0.9 mm) and found shock angle varies non-monotonically with roughness, driven by competing boundary-layer displacement and shock-sampling effects

Variable Pitch Propeller — Design, Build & Test *AS343AI Course Project, AY 2024–25*

- Designed the rack-and-pinion pitch mechanism synchronizing all three blades; 3D-printed the propeller and hub, and wrote Arduino firmware for a custom laser-diode tachometer and BLDC motor speed control
- Validated BEMT/CFD-predicted thrust (peak 3.00 N at 28° pitch) against bench data across 16 pitch configurations, independently cross-checked via actuator-disk momentum theory scripted in Python

Smart Modular Workstation for Disaster Relief *Runner-Up, Vyoma Design-a-thon, May 2025*

- Led CAD design across four independent modules of a 75 kg air-deployable field station and ran static structural FEA for worst-case corner-impact loads, achieving a Factor of Safety of 4.0 at a 30,000 N design load
- Engineered a descent system (parachute + CO_2 airbags) reducing terminal velocity from ~31 m/s to ~6.2 m/s, confirmed both analytically and via CFD

Li-Fi (VLC) — Wireless Optical Communication over Laser *RVCE Experiential Learning, ACY 2023–24*

- Built and wired both hardware generations of a visible-light-communication prototype (LED+LDR, then upgraded to a KY-008 laser) and wrote the Arduino TX/RX firmware handling bit-framing and byte reconstruction
- Wrote a Python encode/decode pipeline enabling image transmission; pushed per-bit transmission from 100 ms to ~1 ms (100x speedup), achieving 360×360px image transfer in ~5 seconds over a free-space optical link

Pneumatic Pressure Measurement & Control System *MATLAB Simulink, AS254IA, AY 2025–26*

- Independently modeled all five subsystems of a pneumatic pressure chain: DC motor/compressor from first-principles piston geometry, tank pressure accumulation as an integrated ODE, comparator-driven relief logic, and 4–20 mA signal conditioning
- Validated sensor output tracking pressure linearly across three operating points (0 bar → 4 mA through 5.07 bar → 20 mA), confirming correct end-to-end signal-chain behaviour

TECHNICAL SKILLS

Software & Tools: ANSYS Fluent | ANSYS Workbench | ANSYS APDL | ANSA | Pointwise | SolidWorks | SolidWorks Flow Simulation | NASTRAN/PATRAN | STK | MATLAB/Simulink | Python (OOP) | Arduino IDE/C++ | LaTeX

Domains & Methods: CFD (incl. multiphase flow) | FEA & structural/flutter analysis | orbital mechanics | gas dynamics & propulsion | 2D/3D mesh generation | DFM | GD&T | CAD | instrumentation & controls | embedded systems

SOFT SKILLS

Team Coordination & Cross-Functional Collaboration | Technical Presentation | Design Thinking | Technical Documentation & Reporting | Workshop/Resource Management

HONORS & AWARDS

- Runner-Up, Vyoma Design-a-thon (May 2025)
- 5th Place, Engineering Design Presentation, Formula Bharat 2026